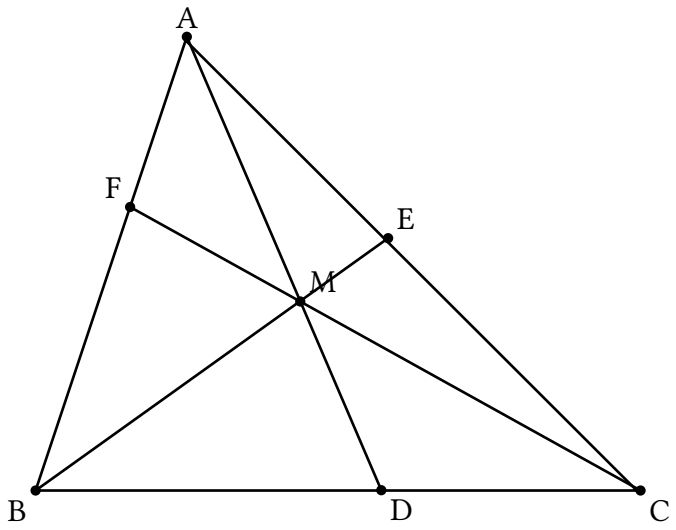
Easy Math Document

Point M inside triangle ABC



Let M be a point inside triangle ABC. The rays AM, BM, CM meet BC, CA, AB at D, E, F respectively.

a) Prove $\frac{MD}{AD} + \frac{ME}{BE} + \frac{MF}{CF} = 1$

Triangles MBC and ABC share the base BC. Since A, M, D are collinear, the distances from M and A to BC are proportional to MD and AD:

$\frac{MD}{AD} = \frac{[MBC]}{[ABC]}$

Similarly:

$\frac{ME}{BE} = \frac{[MCA]}{[ABC]}, \frac{MF}{CF} = \frac{[MAB]}{[ABC]}$

Adding them, since triangles MBC, MCA, MAB exactly compose triangle ABC:

$\frac{MD}{AD} + \frac{ME}{BE} + \frac{MF}{CF} = \frac{[MBC]+[MCA]+[MAB]}{[ABC]} = 1$

b) Prove $\frac{AM}{AD} + \frac{BM}{BE} + \frac{CM}{CF} = 2$

Since M lies between A and D, AM + MD = AD, so:

$\frac{AM}{AD} = 1 - \frac{MD}{AD}$

Similarly for B and C. Adding them and using part a):

$\frac{AM}{AD} + \frac{BM}{BE} + \frac{CM}{CF} = 3 - 1 = 2$

c) At least one ratio ≥ 2 and at least one ratio ≤ 2

Set x = $\frac{MD}{AD}$, y = $\frac{ME}{BE}$, z = $\frac{MF}{CF}$. By part a): x + y + z = 1.

We have:

$\frac{AM}{MD} = \frac{1-x}{x}$, $\frac{BM}{ME} = \frac{1-y}{y}$, $\frac{CM}{MF} = \frac{1-z}{z}$

- Suppose all three ratios are < 2. Then $\frac{1-x}{x} < 2 \Longrightarrow x > \frac{1}{3}$, similarly y > $\frac{1}{3}$ and z > $\frac{1}{3}$. Hence x + y + z > 1, a contradiction.

- Suppose all three ratios are > 2. Then x, y, z < $\frac{1}{3}$, so x + y + z < 1, also a contradiction.

Therefore among the three ratios there is always at least one ratio ≥ 2 and at least one ratio ≤ 2.